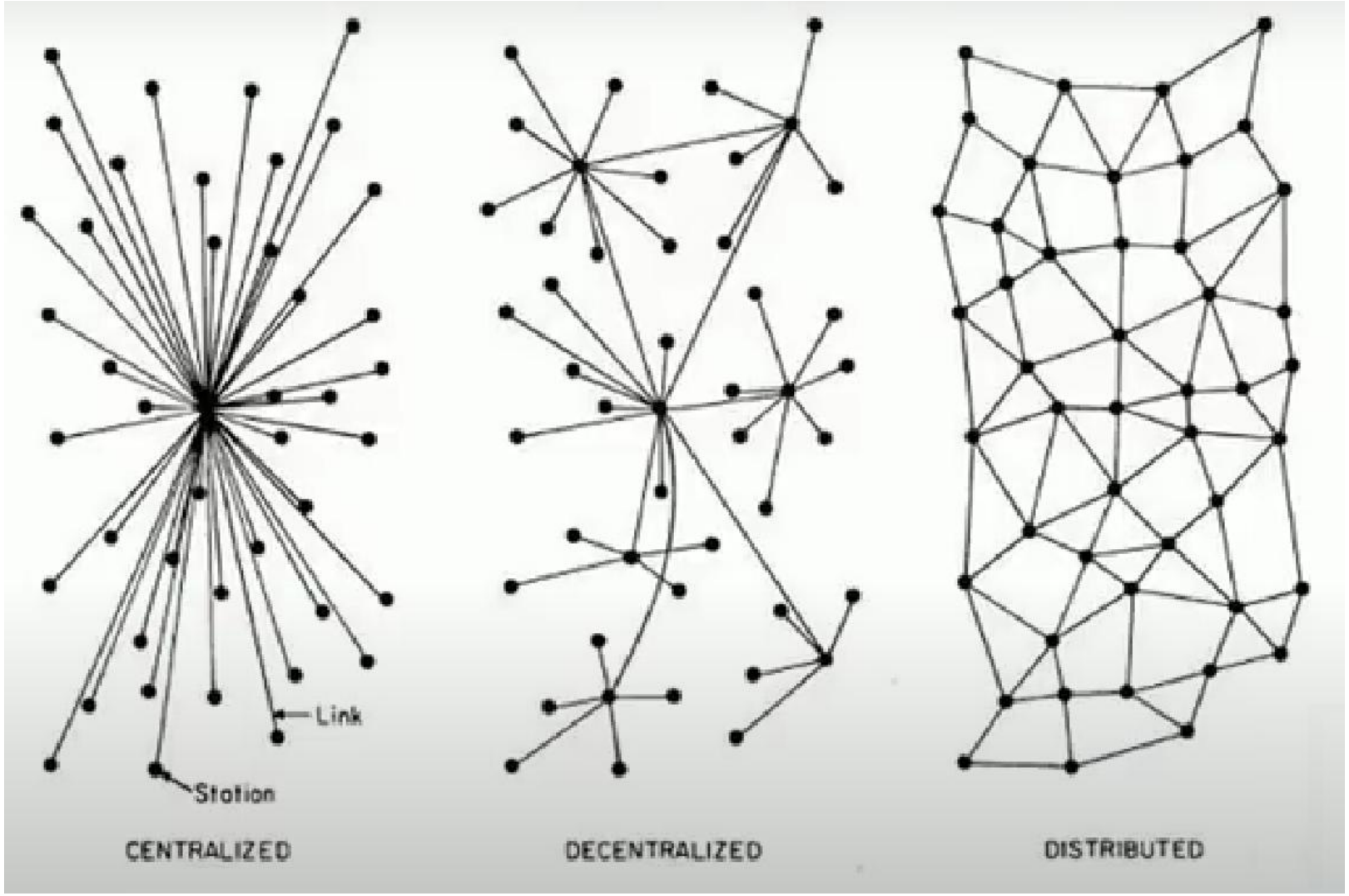
Centralized
Decentralized
Distributed
P2P
Systems

Centralized vs Decentralized vs Distributed

In centralized, complete reliance on single point, that is not safe.

In decentralized, multiple points of coordination.

In distributed system, everyone can collectively execute the job.



Centralized System

- A type of system where control, data processing, and decision-making are concentrated in a single location or entity.
- One central unit (often called the **central server**, **mainframe**, or **central node**) handles most or all of the important tasks, such as data storage, processing, and user interaction.
- Other components (like client machines or peripheral nodes) may only interact with the central system but do not perform major processing themselves.

Features of Centralized System

- **Single Point of Control:** All the decision-making, resource management, and system administration are handled by the central server or unit.
- **Centralized Data Storage:** All data is stored in one central location, which makes it easier to manage and secure.
- **Client-Server Interaction:** In many cases, peripheral units (clients) interact with the central system to request or send data, but they don't handle the actual processing.
- **Simplified Management:** Since there is one main location where tasks are handled, management, monitoring, and troubleshooting can be more straightforward.

Pros and Cons

Ease of Management: With one central point of control, administrators can oversee all system activities.

Consistent Policies: Uniform rules, protocols, and updates can be applied across the entire system.

Simplified Data Backup: Centralized systems make it easier to back up and secure data in one location.

Single Point of Failure: If the central system fails, the entire system could become unavailable.

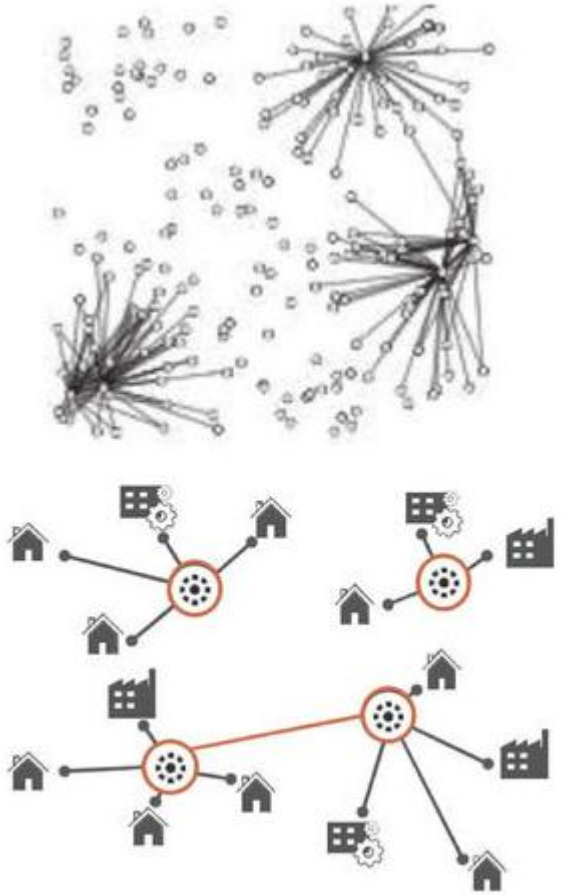
Scalability Issues: As the system grows, the central unit may struggle to handle an increasing volume of requests or data.

Performance Bottlenecks: All data processing and tasks must pass through the central unit, which could cause delays if demand increases.

Examples of Centralized System???

Decentralized System

- A type of system architecture in which control, decision-making, and processing are distributed across multiple independent nodes or entities rather than being concentrated in a single central authority.
- This approach contrasts with **centralized systems**, where a central server or entity holds all the power and responsibility for managing the system's operations.



Features of Decentralized System

- **Distributed Control:** Unlike centralized systems, where one central server or authority governs the entire system, decentralized systems distribute authority and control across multiple nodes or participants.
- **Fault Tolerance:** Since there is no single point of failure, decentralized systems are often more resilient to failures. If one node or participant fails, the system can continue to operate without major disruption.
- **Scalability:** Decentralized systems are often more scalable since new nodes or participants can join and contribute without overwhelming a central server.

Features of Decentralized System

- **Security and Privacy:** In many decentralized systems, data can be distributed or encrypted in a way that makes it more difficult to hack or compromise, providing enhanced security and privacy. Blockchains are a prime example where data is recorded across many nodes, making tampering highly challenging.
- **Autonomy:** Participants in a decentralized system often operate autonomously, making decisions based on their own inputs without needing approval from a central authority.

Advantages of Decentralized System

Resilience: No single point of failure means the system can continue operating even if parts of it go down.

Efficiency: Decentralized systems can be more efficient because they leverage resources across multiple nodes.

Transparency and Trust: In some decentralized systems, like blockchain, transparency is built in, allowing participants to trust the system without a centralized authority.

Disadvantages of Decentralized System

- **Complexity:** Decentralized systems can be more complex to design and manage, especially when it comes to ensuring consensus and coordination between nodes.
- **Data consistency:** Ensuring data consistency across all nodes can be challenging, particularly in real-time or high-speed applications.
- **Security risks:** While decentralization can increase security, it can also introduce new risks, such as vulnerabilities in the communication protocols between nodes.

Use-Cases of Decentralized System

Blockchain: In blockchain networks like Bitcoin and Ethereum, the system is decentralized, with multiple nodes maintaining a shared ledger and agreeing on transactions through consensus mechanisms.

Peer-to-Peer Networks (P2P): File-sharing systems like BitTorrent, where files are distributed directly between users, are decentralized. No central server controls the file storage or transfer.

Decentralized Finance (DeFi): DeFi platforms operate without a central authority, using blockchain technology to allow users to exchange assets, borrow, lend, and trade without traditional intermediaries like banks.

Distributed Databases: Some database systems, such as Apache Cassandra or MongoDB, use decentralized architectures to store data across multiple servers, ensuring no single point of failure.

Distributed System

- A collection of independent computers or nodes that work together to appear as a unified system to users or applications.
- The computers or nodes are connected over a network and coordinate with each other to achieve common goals, such as processing data or providing services.
- Despite the individual nodes, the system presents itself as a single, cohesive entity.

Characteristics Distributed System

- **Multiple Nodes:** A distributed system typically consists of multiple machines or processes, often located in different physical locations.
- **Communication:** These nodes communicate over a network using protocols such as TCP/IP or HTTP.
- **Concurrency:** Each node may perform its own tasks simultaneously, leading to concurrency and the need for synchronization between nodes.
- **Fault Tolerance:** Distributed systems are designed to tolerate failures. Even if one or more nodes fail, the system can continue to operate without significant disruption.

Characteristics Distributed System

- **Scalability:** Distributed systems can scale horizontally (by adding more nodes) to handle increasing workloads.
- **Transparency:** The system should hide the complexity of distributed operations (e.g., data location, network communication) from the user or application, making it appear as though there is a single system.

Types of Distributed Systems

- **Client-Server Systems:** In this setup, clients request services, and servers provide them. This is one of the most common types of distributed system,
- **Peer-to-Peer (P2P) Systems:** Each node in a P2P system is both a client and a server. These systems are decentralized and often used for file-sharing networks like BitTorrent.
- **Cloud Computing:** Cloud platforms such as AWS, Google Cloud, and Azure are distributed systems that offer services, storage, and computing resources across many nodes.
- **Microservices Architecture:** A modern software architecture where a large application is divided into smaller, independent services that communicate over a network

Challenges in Distributed System

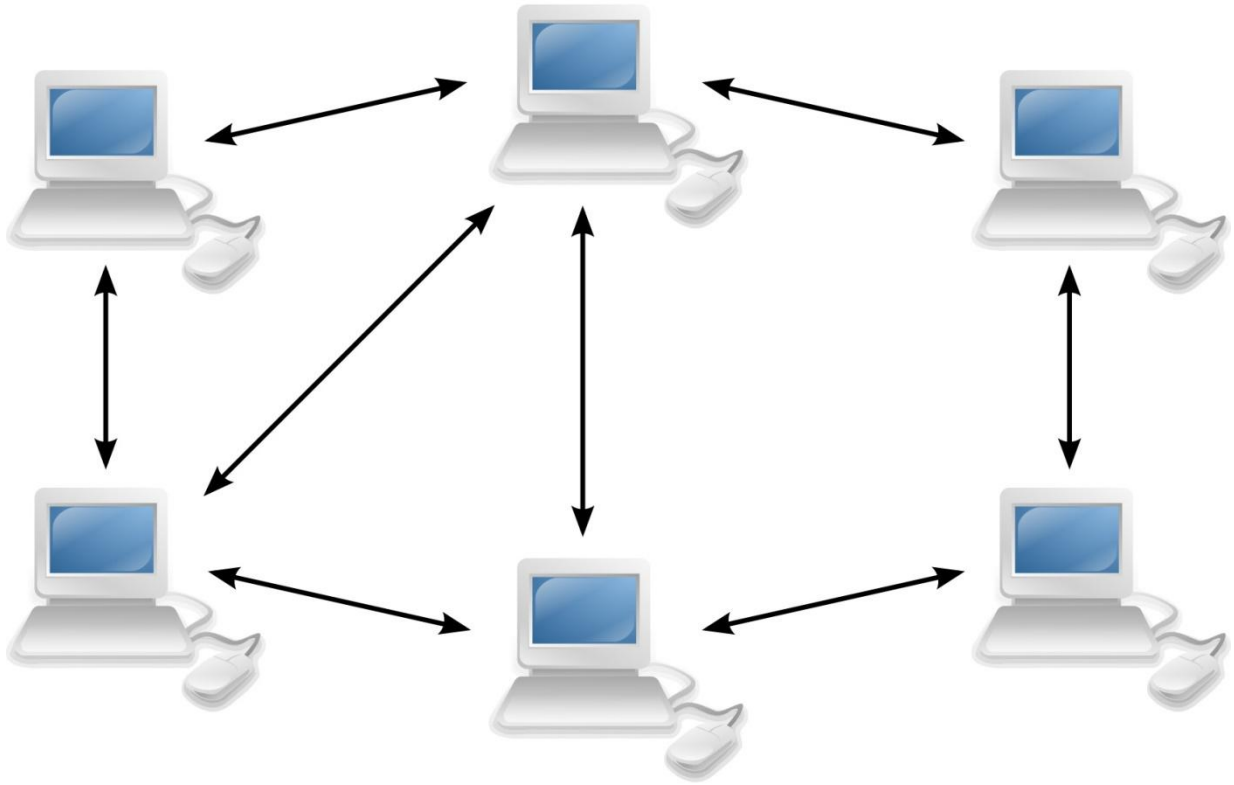
- **Synchronization**: Managing concurrent operations across different nodes and ensuring that they are synchronized.
- **Consistency**: Ensuring that all nodes have a consistent view of the data, especially in cases where the nodes can fail or operate asynchronously.
- **Network Latency**: Communication between nodes can experience delays, which can affect system performance.
- **Fault Tolerance and Recovery**: Managing failures of nodes, network issues, and partial system failures without affecting overall system performance.
- **Security**: Ensuring that communication between nodes is secure and that data is protected across the distributed system.

Use-Cases of Distributed Systems

- **Google Search**: The indexing of billions of web pages happens across multiple servers.
- **Distributed Databases**: Databases such as Cassandra or MongoDB that store data across different machines.
- **Blockchain**: A decentralized distributed ledger where multiple nodes validate transactions.
- **Distributed File Systems**: Systems like Hadoop's HDFS or Google File System (GFS) store and process large data sets across many machines.

P2P Network

- A peer-to-peer (P2P) network is a decentralized type of network architecture where all participants, called nodes or peers, have equal privileges and responsibilities.
- Unlike traditional client-server models, in P2P networks, peers can act as both clients and servers, enabling direct sharing of resources such as files, data, or computational power without the need for a central server.



Characteristics of P2P Networks

- **Decentralization**: There is no central authority or server; all nodes have equal status.
- **Direct Communication**: Nodes communicate directly, allowing resource sharing like files or bandwidth.
- **Scalability**: P2P networks can scale easily as more nodes join the network.
- **Redundancy**: Data is often replicated across multiple nodes, increasing reliability.
- **Cost-Efficiency**: Since resources are shared among peers, there is no need for centralized infrastructure, reducing costs.

Types of P2P Networks

- **Structured P2P:** Nodes are organized in a specific topology, often using a Distributed Hash Table (DHT) for efficient data retrieval.
  - Example: BitTorrent's DHT implementation
- **Unstructured P2P:** Nodes connect randomly without a strict topology, making them easier to set up but less efficient in searching.
  - Example: Gnutella
- **Hybrid P2P:** Combines centralized elements with P2P functionality. A central directory may help peers find each other, but interactions occur peer-to-peer.

Advantages and Disadvantages of P2P Systems

Advantages:

- No single point of failure.
- More efficient use of network resources.
- Greater resilience against censorship or shutdowns.

Disadvantages:

- Harder to secure and manage.
- Vulnerable to malicious peers.
- Potential legal risks, especially in file-sharing contexts.

Use-Cases of P2P Systems

- **File Sharing**: Networks like BitTorrent allow users to share and download large files directly.
- **Blockchain and Cryptocurrencies**: Blockchain technology (e.g., Bitcoin, Ethereum) is inherently P2P, enabling decentralized financial transactions.
- **Communication Tools**: Messaging apps like Skype (older versions) use P2P for voice and video calls.
- **Distributed Computing**: Projects like BOINC (e.g., SETI@home) harness P2P to share computational tasks.